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AI Engineering Leader & Platform Architect

Platform Engineering | Production Applied AI Infrastructure | Distributed Systems at Scale

EXECUTIVE SUMMARY

AI Engineering Leader & Platform Architect with 20+ years building mission-critical distributed systems, launching 0→1 products, and scaling global engineering organizations. Currently leading applied AI infrastructure and platform engineering at Charles Schwab, managing multi-pod teams (~30 engineers), driving production LLM orchestration and inference optimization, and executing zero-downtime microservice migrations at 99.99% availability across systems serving 30M+ users. Track record of shipping high-scale products under hard constraints (1M+ users in 9 months), operationalizing AI workloads on multi-billion dollar transaction platforms, and building engineering cultures that compound velocity over time.

PROFESSIONAL EXPERIENCE

CHARLES SCHWAB | Jersey City, NJ *Senior Technical Lead Manager / Platform & AI Architect* | Oct 2021 – Present

Org Design, Talent & Core Scale

- **Org & Team Scaling:** Built and scaled a high-performing, multi-pod global engineering organization (12 direct reports, ~30 total engineers across US and India GCC hubs), utilizing follow-the-sun execution structures. Established standardized promotion calibrations, defined cross-functional team charters, and reduced core engineer attrition during the massive TDA/Schwab platform integration.
- **Core Modernization & Reliability:** Orchestrated a year-long, zero-downtime infrastructure migration, dismantling a legacy core monolith into 36+ cloud-native microservices. Reduced end-to-end platform transaction latency from 5 seconds to under 2 seconds, maintaining 99.99% system availability while boosting digital user engagement by 18% for 1M+ active equity platform participants.
- **Incident Leadership:** Served as an Incident Commander managing critical high-priority technical incidents; designed fallback runbooks and automated monitoring alerts that safely dropped overall platform MTTR by 35%.

Applied AI Infrastructure

- **Customer-Facing AI Evolution:** Architected the technical migration of the flagship customer-facing conversational digital assistant from a legacy deterministic platform (v1) to an advanced, LLM-powered Semantic Search engine (v2) serving 30M+ active users.

- **AI Infrastructure Optimization:** Engineered prompt guardrails, closed-loop automated evaluation pipelines, and embedding-based retrieval flows using Vertex AI and FAISS; lowered production hallucination rates to near-zero and achieved an estimated 35% automated support ticket deflection.
- **Agentic AI Platforms:** Directed internal developer platform initiatives deploying Copilot MCP agentic integrations and developer automation patterns, streamlining PR review and test consolidation workflows across 15+ engineering teams.
- **FinOps Execution:** Managed multi-region cloud capacity planning across AWS/GCP, utilizing spot instances and optimized vLLM batch structures to slash compute inference infrastructure costs by 22% without breaching platform SLA bounds.

TD AMERITRADE | Jersey City, NJ *Staff Software Engineer / Tech Lead* | 2018 – 2021

- **0→1 Platform Launch (TDx):** Architected and co-delivered the core transactional backend platform for TDx, scaling infrastructure from zero to 1M+ active daily users within 9 months of initial deployment.
- **High-Throughput Systems Architecture:** Owned backend platform systems spanning real-time, sub-second market data ingestion, complex derivative execution pipelines, and high-QPS asset account aggregations operating under strict latency boundaries.
- **Engineering Rigor & CI/CD:** Championed automated testing infrastructure that raised test coverage from 45% to 90%, embedding continuous integration pipelines (GitHub Actions/Jenkins) that safely accelerated deployment frequencies from bi-weekly code freezes to multi-daily code drops.
- **ML Risk Mitigation & SLOs:** Engineered real-time production SLO frameworks and defensive failure containment patterns for automated machine-learning workflows, constructing deterministic fallbacks and explainability layers to protect platform transaction integrity.

MADISON SQUARE GARDEN (MSG) | New York, NY *Senior Software Engineer — Core Data Services* | 2016 – 2018

- **Data Pipeline Orchestration:** Re-engineered legacy data workflows into cloud-native architectures to handle high-volume transactional data, normalizing disparate data streams across global venues to establish a single source of truth.
- **Performance Optimization:** Built high-throughput REST APIs and real-time data ingestion pipelines that reduced reporting latency by 60%, enabling executive leadership to make live, data-driven decisions during high-capacity events.

AXA | Bangalore, India *Technical Lead Manager — Enterprise Cloud Platforms* | 2010 – 2016

- **Org & Team Scaling:** Promoted from Senior Engineer to manage a 10-person global development pod directing the strategic delivery of enterprise-scale BI and regulatory reporting platforms for 8+ European markets.
- **Data Platform Migration:** Spearheaded migration of core sales infrastructure to SAP HANA, migrating 10+ TB of sales data, streamlining global data pipelines, and reducing cross-border reporting cycle times by 45%.

- **Process Maturity:** Standardized automated delivery architectures and engineering governance practices across global pods, directly securing the organization's CMMI Level 3 certification and receiving the organization's CEO Award for engineering excellence.

SAP LABS | Bangalore, India *Senior Software Engineer — Release Automation Infrastructure* | 2005 – 2010

- **Release Automation & Orchestration:** Built an automated release orchestration engine handling complex multi-version deployments, expanding global release velocity across the enterprise ecosystem.
- **Core Systems Architecture:** Contributed core foundational architecture to the Sydney Enterprise Services Project, earning the *SAP Kernel Assembly Team Award* for architectural contributions that became the standard release framework for the PLM product line.

SELECTED PROJECTS

- **Careerflow360:** Agentic AI college recommendation engine implementing RAG and multi-step LLM orchestration using LangGraph and ChromaDB.
- **GitPeek:** Enterprise secure code governance tool utilizing OAuth-based zero-trust repository access controls.

TECHNICAL SKILLS

- **Applied AI & Inference Platforms:** LLM Orchestration (vLLM, LangGraph, CrewAI, LangSmith), Vector Databases (FAISS, ChromaDB, Pinecone), Enterprise RAG Architecture, Automated Evaluation Loops, Embedding Pipelines, Vertex AI.
- **Distributed Platforms & Cloud Core:** Cloud Architecture (AWS, GCP), Core Compute & Storage (EC2, S3, GCS), High-QPS Microservices, Event-Driven Streaming (Kafka), In-Memory Caching (Redis), Relational/NoSQL Databases (PostgreSQL, MongoDB, SQL, GraphQL, REST APIs).
- **Containerization, IaC & Observability:** Docker, Kubernetes, Terraform, Continuous Integration & Rigor (CI/CD, GitHub Actions, Jenkins), Test Automation (Selenium, Appium, SauceLabs), Observability & APM (Datadog, Splunk).
- **Languages & Frameworks:** Python (FastAPI), Node.js, TypeScript, JavaScript, Java, C, C++, Rust.

EDUCATION & PUBLICATIONS

- MTech, Information Technology (Specialization: Intelligent Systems & Robotics), [IIIT Allahabad](#), INDIA
- BTech, Computer Engineering, [Pantnagar University](#), INDIA
- Publication: “[AI-driven zero-moment point simulation in robotics](#)” - Carnegie Mellon University (CMU), 2007